

You will be familiar with this one rule, and most likely remember the name too!

Chain rule

$$h(x) = f(g(x))$$

○

It says that if we consider a function which is applied to another function

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) =$$

○

Then we can compute its derivative in terms of the two functions that it is made up of

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

○

Notice the the derivative of the outer is applied to the inner function

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

○

This has an important meaning in our case as you will see

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$h(x) = \sin(x^2)$$

But first let us make sure you are clear about the procedure

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$h(x) = \sin(x^2)$$

$$h'(x) = \cos(x^2).2x$$

You can see that the inner function was x^2 and so cosine is applied to x^2

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$h(x) = \sin(x^2)$$

$$h'(x) = \cos(x^2).2x$$

But we also multiply by the derivative of x^2

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

Let us consider the example of a parametrization

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t))$$

How does a reparametrization affect the velocity vector?

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

Remember that this time our function has two coordinates, so we must find the rate of change of each

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t))$$

We apply chain rule. Check to confirm that this is indeed what we get

Chain rule

$$h(x) = f(g(x))$$

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But what does this mean geometrically?

Chain rule

$$h(x) = f(g(x))$$

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Chain rule

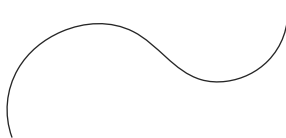
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The parametrization traces out a curve

Chain rule

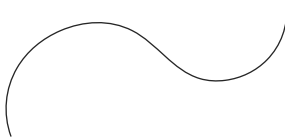
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Chain rule

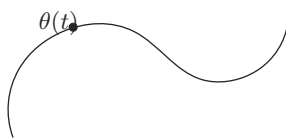
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$\theta(t)$ is some point on the curve

Chain rule

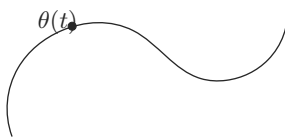
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Chain rule

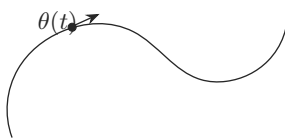
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And its derivative is simply some velocity vector at that point

Chain rule

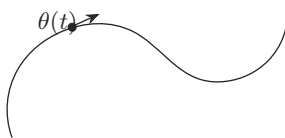
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But what happens if we change the parametrization by τ ?

Chain rule

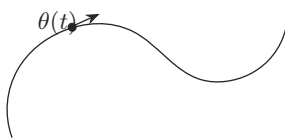
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Chain rule

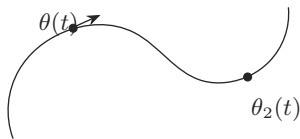
$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

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$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$



Remember that $\theta_2(t)$ not is the same point as $\theta(t)$

Chain rule

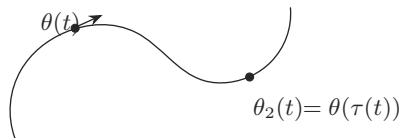
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$\theta_2(t)$ is the same point as $\theta(\tau(t))$

Chain rule

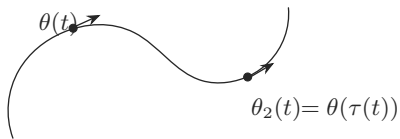
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$$\theta(t) = (f(t), g(t))$$

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If this is its velocity vector using θ

Chain rule

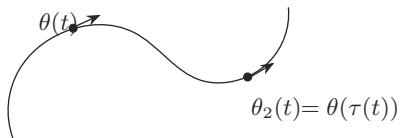
$$h(x) = f(g(x))$$

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$$\theta(t) = (f(t), g(t))$$

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Chain rule

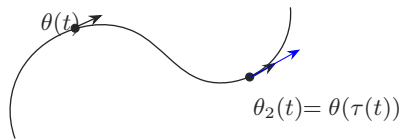
$$h(x) = f(g(x))$$

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$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$



The new velocity vector is scaled up because observe that each coordinate is multiplied by the same $\tau'(t)$

Chain rule

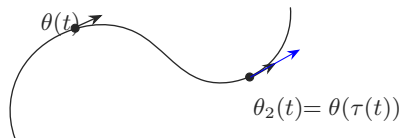
$$h(x) = f(g(x))$$

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$$\theta(t) = (f(t), g(t))$$

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Let us now consider a simple but explicit example

Chain rule

$$h(x) = f(g(x))$$

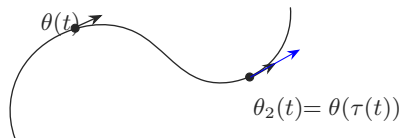
$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example



This will be our recurring example to test our theories

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

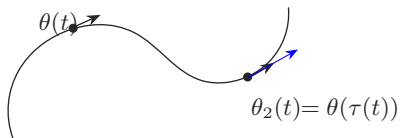
$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$



So once again we consider the circle parametrized as shown

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

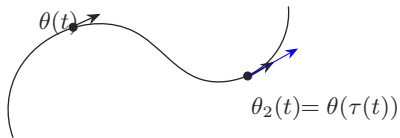
$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$



We can easily calculate its velocity for any t

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

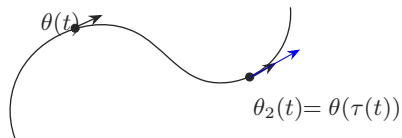
$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$



Which means that for the point $\theta(\pi/4)$, this is the velocity vector

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

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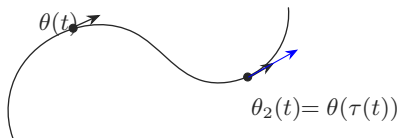
Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$



But let us say that we reparametrize it by doubling the speed

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

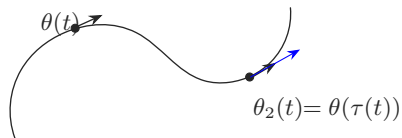
$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t)$$



This is the new parametrization

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

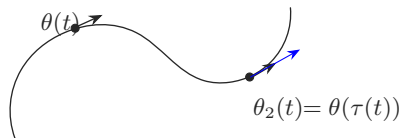
$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$



which is then explicitly written as this

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

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Example

$$\theta(t) = (\cos(t), \sin(t))$$

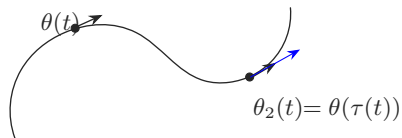
$$\theta'(t) = (-\sin(t), \cos(t))$$

$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$



Now we apply chain rule on the derivative

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

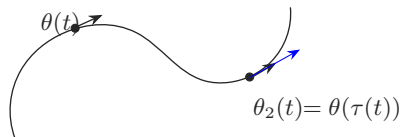
$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



But now the same point is $\theta(\pi/8)$ not at $\pi/4$

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

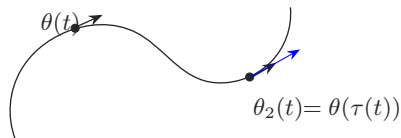
$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Luckily, the chain rule took care of that!

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

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Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

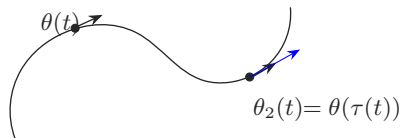
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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



And we find that the velocity is now double of the previous one!

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

$$\theta(t) = (f(t), g(t))$$

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$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

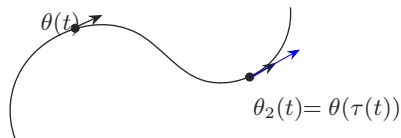
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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



But very crucially, the direction has remained unchanged

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

1. $h^{-1}(h(x)) = x$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

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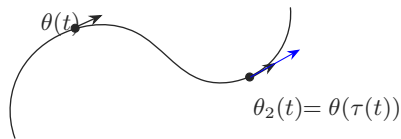
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$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Let us now apply the chain rule to the inverse of a function

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

1. $h^{-1}(h(x)) = x$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

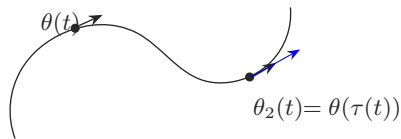
$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



That means two different things

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

1. $h^{-1}(h(x)) = x$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

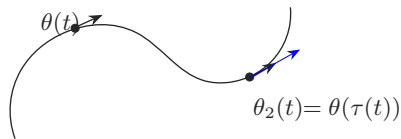
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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



One that applying the inverse to the original gives back the same thing

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

1. $h^{-1}(h(x)) = x$

2. $h(h^{-1}(x)) = x$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta'_2(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

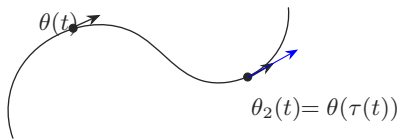
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$$\theta'_2(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta'_2(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



But also, that applying the original to the inverse gives back the same thing

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \ h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$2. \ h(h^{-1}(x)) = x$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

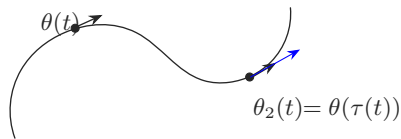
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$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Applying chain rule, we can get a nice relation for the derivative of the inverse in terms of the original

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \ h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \ h(h^{-1}(x)) = x$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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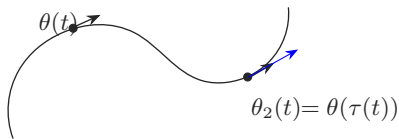
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Notice that the derivative of the inverse is applied to $h(x)$ and not x itself.

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \ h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \ h(h^{-1}(x)) = x$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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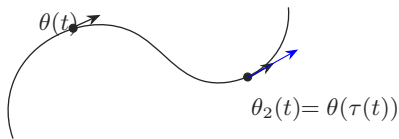
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



That is simply because x in the domain of h corresponds to the $h(x)$ in the domain of the inverse

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \ h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \ h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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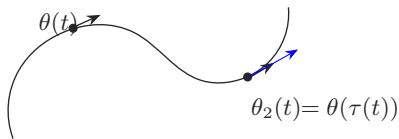
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



When we do a similar thing with the other expression

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

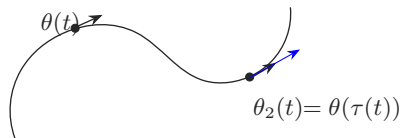
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$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



we get the same relation but with one small difference

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

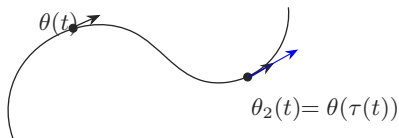
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Earlier we were applying the derivative of the inverse on $h(x)$ so it was in terms...

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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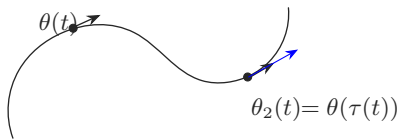
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



The derivative of the original applied to the inverse of $h(x)$ which is x

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

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Example

$$\theta(t) = (\cos(t), \sin(t))$$

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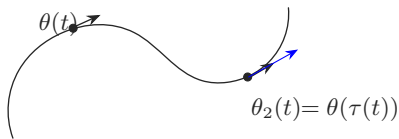
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



But now we are applying the derivative of the inverse to x itself

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

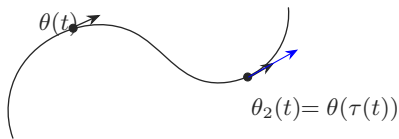
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



and its inverse is $h^{-1}(x)$ that is why this time we see the derivative of h applied to $h^{-1}(x)$

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

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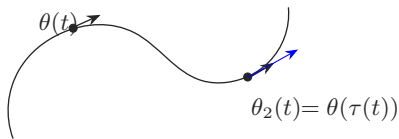
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$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



You would know this theorem but may not know it by this name

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) =$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

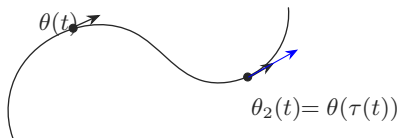
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



Let us consider a function that is defined differently

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

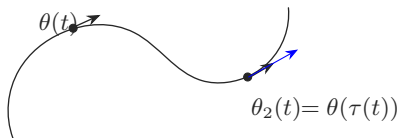
$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



by using an integral. Notice that it only the upper end point of the interval that is a variable

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

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The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

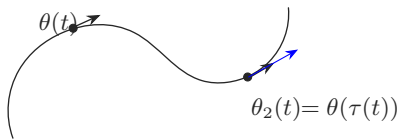
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$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



The rest are fixed

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

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$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

$$\theta(t) = (f(t), g(t))$$

$$\theta_2(t) = \theta(\tau(t)) = (f(\tau(t)), g(\tau(t)))$$

$$\theta_2'(t) = (f'(\tau(t))\tau'(t), g'(\tau(t))\tau'(t)) = \theta'(\tau(t))\tau'(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t))$$

$$\theta'(t) = (-\sin(t), \cos(t))$$

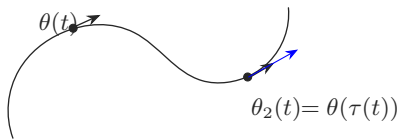
$$\theta'(\pi/4) = (-1/\sqrt{2}, 1/\sqrt{2})$$

$$\tau(t) = 2t$$

$$\theta_2(t) = \theta(2t) = (\cos(2t), \sin(2t))$$

$$\theta_2'(t) = (-\sin(2t)2, \cos(2t)2)$$

$$\theta_2'(\pi/8) = (-\sqrt{2}, \sqrt{2})$$



This theorem says that the derivative of the integral is the function we were integrating

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

Let $f(t)$ denote the speed at time t

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

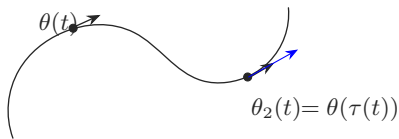
$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$



Keep this example in mind to see why it is so

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

Let $f(t)$ denote the speed at time t

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

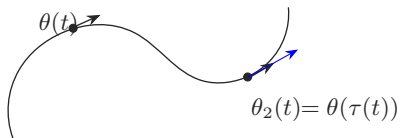
$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$



A function little f can always represent the speed of a car at time t

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

Let $f(t)$ denote the speed at time t

$$F(x) = \int_a^x f(t)dt$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

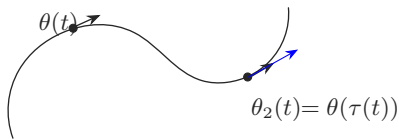
$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$



Do you see what this integral should mean in this context?

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

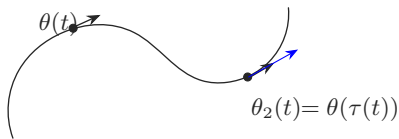
The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x



The integral shown here is the distance travelled until time x

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

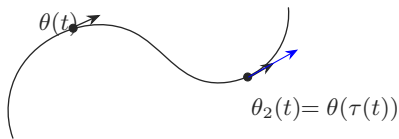
$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x



But we know that differentiating the distance gives the derivative

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

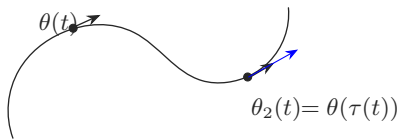
$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again



And that is why we get back x

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Let us now apply all this in figuring out the curvature

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let us consider a parametrization which is not necessarily unit speed

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$

and we can make it unit speed

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

by using the arc length function, as discussed the last time

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s)$$

Let us confirm that it is unit speed by using chain rule to find out the derivative

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

We are computing the derivative of $L^{-1}(s)$ using the highlighted formula

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s))$$

Now this part of the expression is already worked out

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s)) \frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s)) \frac{1}{\|\theta'(L^{-1}(s))\|}$$

But now we can use the fundamental theorem of calculus to finally figure out the derivative of L too!

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s)) \frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s))\frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

$$\|\theta_2'(s)\| = \frac{\|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|}$$

Notice that this shows us that θ_2 that we just obtained

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s))\frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

$$\|\theta_2'(s)\| = \frac{\|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|} = 1$$

is unit speed

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s)) \frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

$$\|\theta_2'(s)\| = \frac{\|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|} = 1$$

$$\theta_2''(s) = \left(\frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|} \right)'$$

We now begin the computation of the second derivative

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s))\frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

$$\|\theta_2'(s)\| = \frac{\|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|} = 1$$

$$\theta_2''(s) = \left(\frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|} \right)'$$

$$= \frac{\|\theta'(L^{-1}(s))\|\theta''(L^{-1}(s)) - \theta'(L^{-1}(s))\|\theta'(L^{-1}(s))\|'}{\|\theta'(L^{-1}(s))\|^2}$$

This will need the quotient rule of differentiation

Chain rule

$$h(x) = f(g(x))$$

$$h'(x) = h'(g(x))g'(x)$$

If h^{-1} is the inverse of h ,

$$1. \quad h^{-1}(h(x)) = x$$

$$h^{-1'}(h(x))h'(x) = 1$$

$$h^{-1'}(h(x)) = \frac{1}{h'(x)}$$

$$2. \quad h(h^{-1}(x)) = x$$

$$h'(h^{-1}(x))(h^{-1})'(x) = 1$$

$$h^{-1'}(x) = \frac{1}{h'(h^{-1}(x))}$$

The Fundamental Theorem of Calculus

$$F(x) = \int_a^x f(t)dt$$

$$F'(x) = f(x)$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\|du$

$$\theta_2'(s) = \theta'(L^{-1}(s))L^{-1'}(s) = \theta'(L^{-1}(s))\frac{1}{L'(L^{-1}(s))}$$

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which says that the denominator is the square of the original denominator

Chain rule

$$h(x) = f(g(x))$$

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While the numerator has the original denominator...

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times the derivative of the numerator, one which we have to apply the chain rule by first differentiating the outside

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and we subtract the numerator times the derivative of the denominator

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But now we are faced with a derivative that will need some work to unravel

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Let us examine the meaning of the norm

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It will be much easier to differentiate if we square and use implicit differentiation

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If $(x_1, y_1) \cdot (x_2, y_2) = 0$, then $\cos(\theta_1 - \theta_2) = 0$, i.e.
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A property of unit speed parametrizations

$$\theta(t) = (x(t), y(t))$$

$$\|\theta'(t)\| = c$$

$$\frac{d}{dt} \|\theta'(t)\| = 0$$

Even better:

$$\|\theta'(t)\|^2 = c^2$$

$$x'^2(t) + y'^2(t) = c^2$$

$$2(x'(t)x''(t) + y'(t)y''(t)) = 0$$

$$(x'(t), y'(t)) \cdot (x''(t), y''(t)) = 0$$

$$\theta'(t) \cdot \theta''(t) = 0$$

Let $f(t)$ denote the speed at time t

$F(x) = \int_a^x f(t)dt$ is the distance covered until time x

So $F'(x)$ must be the speed at x which is $f(x)$ back again

Given a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$

Let $\theta_2(s) = \theta(L^{-1}(s))$ where $L(s) = \int_a^s \|\theta'(u)\| du$

$$\theta_2'(s) = \theta'(L^{-1}(s)) L^{-1}'(s) = \theta'(L^{-1}(s)) \frac{1}{L'(L^{-1}(s))}$$

$$= \theta'(L^{-1}(s)) \frac{1}{\|\theta'(L^{-1}(s))\|} = \frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|}$$

$$\|\theta_2'(s)\| = \frac{\|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|} = 1$$

$$\theta_2''(s) = \left(\frac{\theta'(L^{-1}(s))}{\|\theta'(L^{-1}(s))\|} \right)'$$

$$= \frac{\|\theta'(L^{-1}(s))\| \theta''(L^{-1}(s)) - \theta'(L^{-1}(s)) \|\theta'(L^{-1}(s))\|'}{\|\theta'(L^{-1}(s))\|^2}$$

$$= \frac{\theta''(L^{-1}(s)) - \theta'(L^{-1}(s)) (\theta'(L^{-1}(s)) \cdot \theta'(L^{-1}(s))) / \|\theta'(L^{-1}(s))\|}{\|\theta'(L^{-1}(s))\|^2}$$

$\theta(t) = \theta_2(L(t))$ where θ_2 is unit speed

Exercise Use this relation to derive the same formula for the curvature

$$(x_1, y_1) \cdot (x_2, y_2) := x_1 y_1 + x_2 y_2$$

In polar coordinates,

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For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

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$$\begin{aligned} \theta(t) &= (x(t), y(t)) \\ \|\theta'(t)\| &= c \\ \frac{d}{dt} \|\theta'(t)\| &= 0 \end{aligned}$$

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$$\begin{aligned} \|\theta'(t)\|^2 &= c^2 \\ x'^2(t) + y'^2(t) &= c^2 \\ 2(x'(t)x''(t) + y'(t)y''(t)) &= 0 \\ (x'(t), y'(t)) \cdot (x''(t), y''(t)) &= 0 \\ \theta'(t) \cdot \theta''(t) &= 0 \end{aligned}$$

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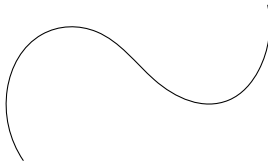
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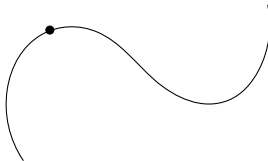
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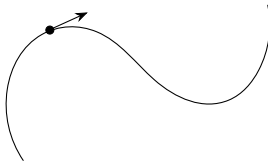
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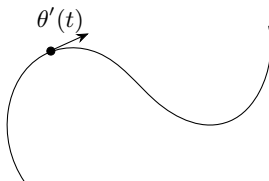
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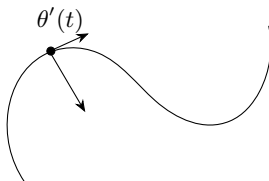
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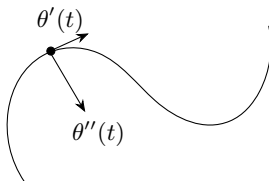
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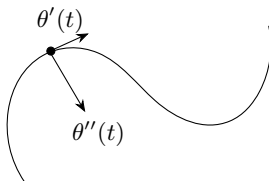
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A constant curvature curve is a circle



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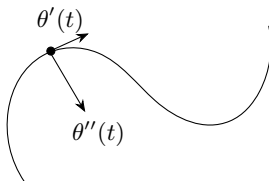
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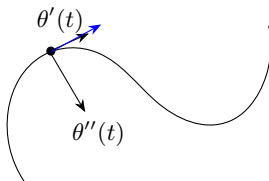
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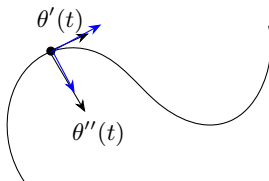
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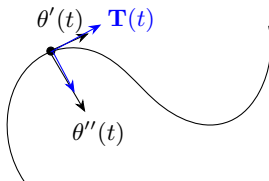
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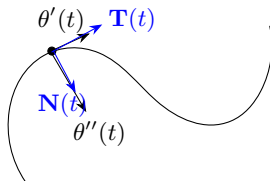
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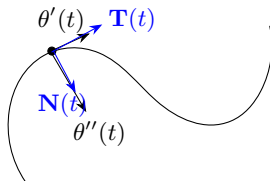
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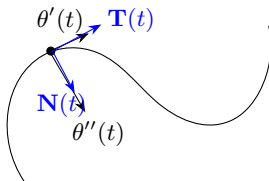
$$(x'(t), y'(t)) \cdot (x''(t), y''(t)) = 0$$

$$\theta'(t) \cdot \theta''(t) = 0$$

For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

$\theta(t)$ is unit speed



$$(x_1, y_1) \cdot (x_2, y_2) := x_1 y_1 + x_2 y_2$$

In polar coordinates,

$$\begin{aligned} (r_1 \cos(\theta_1), r_1 \sin(\theta_1)) \cdot (r_2 \cos(\theta_2), r_2 \sin(\theta_2)) \\ = r_1 r_2 \cos(\theta_1) \cos(\theta_2) + r_1 r_2 \sin(\theta_1) \sin(\theta_2) \\ = r_1 r_2 \cos(\theta_1 - \theta_2) \end{aligned}$$

If $(x_1, y_1) \cdot (x_2, y_2) = 0$, then $\cos(\theta_1 - \theta_2) = 0$, i.e.
 $\theta_1 - \theta_2 = \pi/2$

A property of unit speed parametrizations

$$\theta(t) = (x(t), y(t))$$

$$\|\theta'(t)\| = c$$

$$\frac{d}{dt} \|\theta'(t)\| = 0$$

Even better:

$$\|\theta'(t)\|^2 = c^2$$

$$x'^2(t) + y'^2(t) = c^2$$

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$$(x'(t), y'(t)) \cdot (x''(t), y''(t)) = 0$$

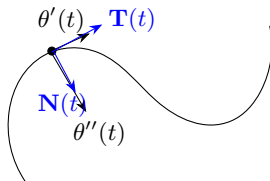
$$\theta'(t) \cdot \theta''(t) = 0$$

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$\theta(t)$ is unit speed

$$\mathbf{T}(t) = \theta'(t)$$



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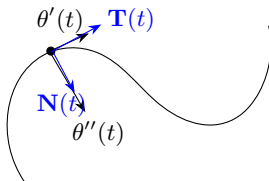
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$$\mathbf{T}(t) = \theta'(t)$$

Let $\mathbf{N}(t)$ denote the unit vector obtained by rotating $\mathbf{T}(t)$ clockwise



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A constant curvature curve is a circle

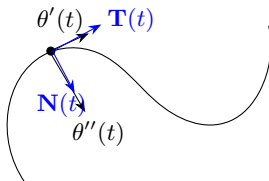
$\theta(t)$ is unit speed

$$\mathbf{T}(t) = \theta'(t)$$

Let $\mathbf{N}(t)$ denote the unit vector obtained by rotating

$\mathbf{T}(t)$ clockwise

$$\theta''(t) = \kappa \mathbf{N}(t)$$



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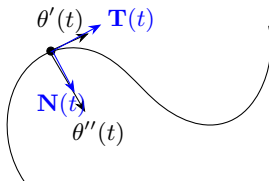
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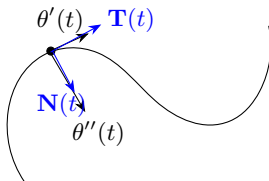
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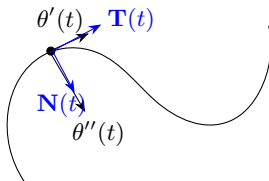
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A constant curvature curve is a circle

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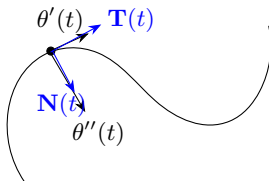
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$$C'(t) = \theta'(t) + \frac{1}{\kappa} \mathbf{N}'(t)$$



$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

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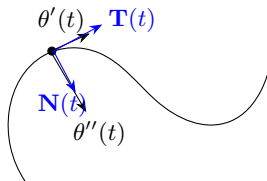
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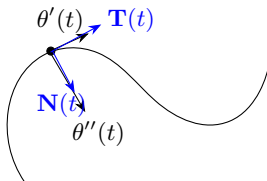
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A constant curvature curve is a circle

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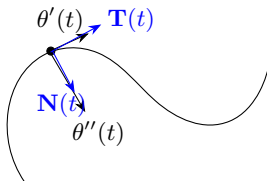
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$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \kappa(t) \mathbf{N}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \kappa(t) = 0$$

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A constant curvature curve is a circle

$\theta(t)$ is unit speed

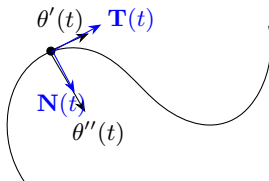
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$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

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If $\mathbf{v}(t)$ is a unit vector

For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

$\theta(t)$ is unit speed

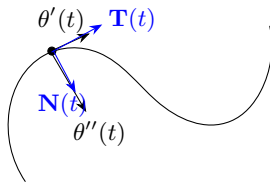
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If $\mathbf{v}(t)$ is a unit vector

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A constant curvature curve is a circle

$\theta(t)$ is unit speed

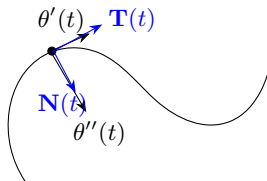
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$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \kappa(t) = 0$$

If $\mathbf{v}(t)$ is a unit vector

$$\mathbf{v}(t) \cdot \mathbf{v}(t) = 1$$

$$2\mathbf{v}'(t) \cdot \mathbf{v}(t)$$

For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

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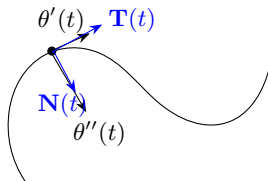
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For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

$\theta(t)$ is unit speed

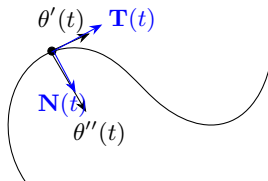
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$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \kappa(t) = 0$$

If $\mathbf{v}(t)$ is a unit vector

$$\mathbf{v}(t) \cdot \mathbf{v}(t) = 1$$

$$2\mathbf{v}'(t) \cdot \mathbf{v}(t) = 0$$

So $\mathbf{v}'(t)$ is perpendicular to $\mathbf{v}(t)$

For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

$\theta(t)$ is unit speed

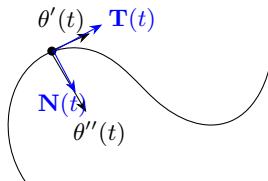
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So $\mathbf{v}'(t)$ is perpendicular to $\mathbf{v}(t)$

So $\mathbf{N}'(t)$ must be perpendicular to $\mathbf{N}(t)$

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$\theta(t)$ is unit speed

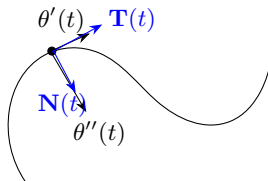
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$\theta''(t) = \kappa \mathbf{N}(t)$ where κ is the constant curvature

$C(t) = \theta(t) + \frac{1}{\kappa} \mathbf{N}(t)$ we hope that $C(t)$ is a constant

$$C'(t) = \theta'(t) + \frac{1}{\kappa} \mathbf{N}'(t)$$



$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

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So $\mathbf{v}'(t)$ is perpendicular to $\mathbf{v}(t)$

So $\mathbf{N}'(t)$ must be perpendicular to $\mathbf{N}(t)$

and must be a scalar multiple of $\mathbf{T}(t)$

For a constant speed parametrization, the acceleration vector is always 90 degrees to the velocity vector

A constant curvature curve is a circle

$\theta(t)$ is unit speed

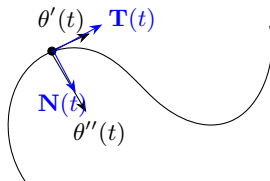
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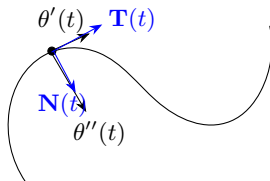
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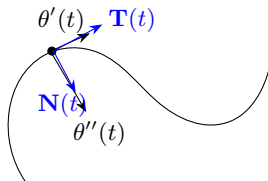
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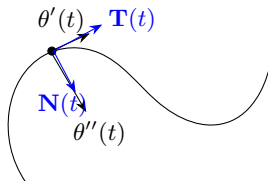
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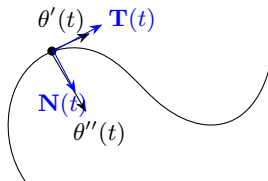
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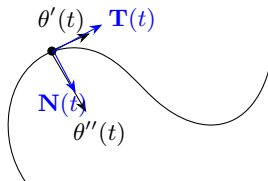
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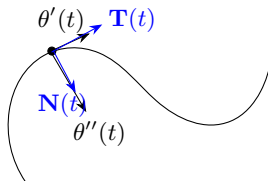
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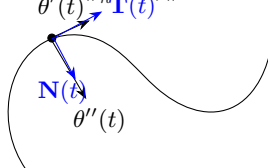
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